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Experimental Investigation of a Novel Morphing Wing Design

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The aerodynamic performance of a novel morphing wing design that actuates its skin to change its effective camber was studied experimentally and numerically. The morphing wing was designed with an Eppler 479 airfoil. Force-based experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) to compare the performance between the morphing wing and a traditional wing with the effective camber deflection between 0° and 5.1° under an angle of attack range of -15° and 15 °. The experiments were conducted at the Reynolds number of 270,000. A panel and lattice method based software called FlightStream were used for the numerical simulation of the cases and were compared with the experimental result. Both experimental and numerical results indicate similar lift performance of the traditional and morphing wing at the same effective camber change and the total drag was comparatively lower for the morphing wing case.

Nomenclature

AR	=	Aspect Ratio
C_D	=	Drag Coefficient, $C_D = \frac{D}{qS_{Ref}}$
C_{D_0}	=	Drag Coefficient, $C_D = \frac{D}{qS_{Ref}}$
C_L	=	Lift Coefficient, $C_D = \frac{L}{qS_{Ref}}$
C_M	=	Moment Coefficient, $C_M = \frac{M}{qS_{Ref}}$
C_P	=	Pressure Coefficient
D	=	Drag force, (N)
L	=	Lift force, (N)
M	=	Pitching Moment, $(N.m)$
q	=	Dynamic Pressure, (Pa)
Re	=	Reynolds Number
S_{ref}	=	Reference area, (m^2)
V_∞	=	Freestream Velocity, (m/s)
α	=	Angle of Attack, (Deg)
ρ	=	Air density, (kg/m^3)
δ_{eff}	=	Effective Camber Angle, (deg)

I. Introduction

The concept of the morphing wing has been around for decades, including wing twisting, wing sweeping, wing folding, span extending, and camber changing [1]. Recently, the interest in camber-changing morphing wings has increased with the advent of fixed-wing Unmanned Aerial Vehicles (UAV). Compared to a traditional trailing edge hinged device, that causes severe drag penalities due to the gaps between the wing body and the trailing edge device, the morphing wing has a smooth wing surface transition along with a continuous change in the wing camber, providing a better aerodynamic performance. Many morphing wing designs has been investigated analytically, experimentally, and computationally since the 1980s [2,4,7,11-14]. Applying the morphing wing concept on modern fixed-wing UAVs would provide better aerodynamic performance, hence, a longer range for drone operation.

To achieve effective camber change, the wing airfoil shape can be morphed either at the leading edge or the trailing edge. Sodja et al. [3] designed and studied the structural performance of a morphing leading-edge design. Due to the small radius at the wing's leading edge, the morphing structure had challenges due to skin bulking. In morphing wings, the trailing edge morphing is more common than the leading edge. Henry et al. [4] designed and studied the structural performance of a morphing wing using MFC piezoelectric patches providing a smooth morphing structure and a better

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overall stiffness. Bilgen et al. [5] designed and built a 1.04m wing span UAV using the piezoelectric morphing wing, a sufficient roll rate was achieved using the morphing wing, however, the roll rate is ~45% lower than the traditional aileron actuation on the same aircraft. Vasista and Tong [6] designed a morphing structure with pressurized chord-wise cells. In this design, morphing is achieved due to the difference in pressure at the airfoil's upper and lower side causing the cell walls to buckle and deform. However, the changes in the camberline due to morphing were limited in the study.

Modern morphing wing desgins are usually made of a combination of rigid and soft materials. Yokozeki and Sugiura [7], Schlup et al. [8] and Navaratne et al. [9] designed a morphing wing internal structure using corrugated shapes to avoid skin bulking. However, the skin design for such morphing structures had a few issues such as a discontinuation at the lower surface of the wing to prevent skin bulking. This led to an increment in the drag force of the wing [7]. Pankonien et al. [10] designed and built a more complicated morphing wing using a macro-fiber composite where the trailing edge was actuated by metal wiring and a hinge system. A silicon skin was used to provide a smooth surface morphing during the actuation process. However, the durability of the silicon skin and the strength of the actuation mechanism became a major drawback of the study. Mkhoyan et al. [11] designed and manufactured a glass fiber morphing wing with 6 individual morphing sections along the span. A set of linear actuators were located at 60% chord location that morphs the trailing edge. Silicon fillings were used to fill the gaps between each morphing section to provide a smooth morphing.

Bio-inspired fish-bone morphing concepts are more successful in recent years, especially with the study conducted by Rivero et al. [12-14] where two design iterations were built and studied. For the first iteration, the wing structure had fish-bone shape stringers at 40% chord location, a solid bending beam spline, soft ‘tendons’ materials for actuation, and elastomeric matrix composite material skin. Even though the wing morphing was achieved, it wasn’t stiff enough for wind tunnel investigations. For the second design iteration, the morphing location was moved back to 75% chord while more solid material is used for the rigid wing part, with a very similar actuation method. The NACA 23012 airfoil was used for the model with a span of 1 m and a chord of 0.27 m. A change in lift coefficient of 0.7 was achieved with the wing morphing. When compared to a traditional flap configuration, the fishbone morphing wing showed a reduction in drag force reduction up to 70% when generating a similar lift force resulting in a higher aerodynamic efficiency. Particle Image Velocimetry (PIV) results revealed a much smaller wake size for the morphing wing resulting in a smaller momentum deficit.

Except for the piezoelectric morphing wing concept, none of the other morphing wing designs was able to achieve camber morphing using skin actuation and simple mechanisms. For a smaller-size fixed-wing UAV, a morphing wing with a simpler mechanical design capable of providing better performance than the traditional trailing edge device is needed to increase the operational capability of the vehicle. The current study proposes a novel morphing wing design using skin actuation and simple structural design which can be implemented in a small fixed-wing UAV.

II. Experimental setup

All experiments were conducted in the University of Dayton’s Low Speed Wind Tunnel (UD-LSWT) in its open jet configuration. The tunnel has a 16:1 contraction ratio, six anti-turbulence screens, and a 0.762m by 0.762m open jet test section. The effective length of the test section is 0.183m. A 1.128m by 1.128m collector collects the expanded air on to the diffuser. A pitot tube is connected to a TSI T600 Micromanometer which was used to measure the freestream velocity during the experiment.

A. Experimental Setup

The force-based experimental setup is shown in Figure 1. The wing model was mounted to a rotary stage and an ATI Gamma F/T [14] balance assembly. A splitter plate of 0.914m x 1.829m was used to obtain a half-span configuration of the models tested. The testing matrix of the experiment is shown in Table 1, the freestream velocity for the morphing wing case was set at 15 and 20 m/s, corresponding to a chord-based Re of 200,000 and 270,000. The angle of attack was varied from -15° to 15° with 1° increment. The data sampling rate was 1000 Hz, for a 15-second duration at each angle of attack for each case, with two runs to improve data repeatability for each data point. The uncertainty of the sensor is 1.25% on its Y-axis, which measures the normal force of the wing, and 0.75% on its X-axis, which measures the axial force of the wing.

The morphing wing was designed using the Eppler 479 airfoil and has a span of 0.406m and a chord of 0.203m. In general, the design has a rigid leading-edge section built of PLA plastic and a flexible trailing edge which is also connected to the skin of the wing built by Varioshore TPU material. Four AGFRC B13DLM servo motors are used to actuate the morphing wing. Three different servo deflection angles: neutral, half-travel, and full-travel are tested, which is controlled by a Arduino Uno microcontroller board. The airfoil shape of the morphing wing is shown in Figure 2. The effective change in the camber angle, δ_{eff}, which is measured between the centerline of the neutral position to the leading-trailing edge line for the morphing wing, are 0°, 3.0°, and 5.1°, respectively.

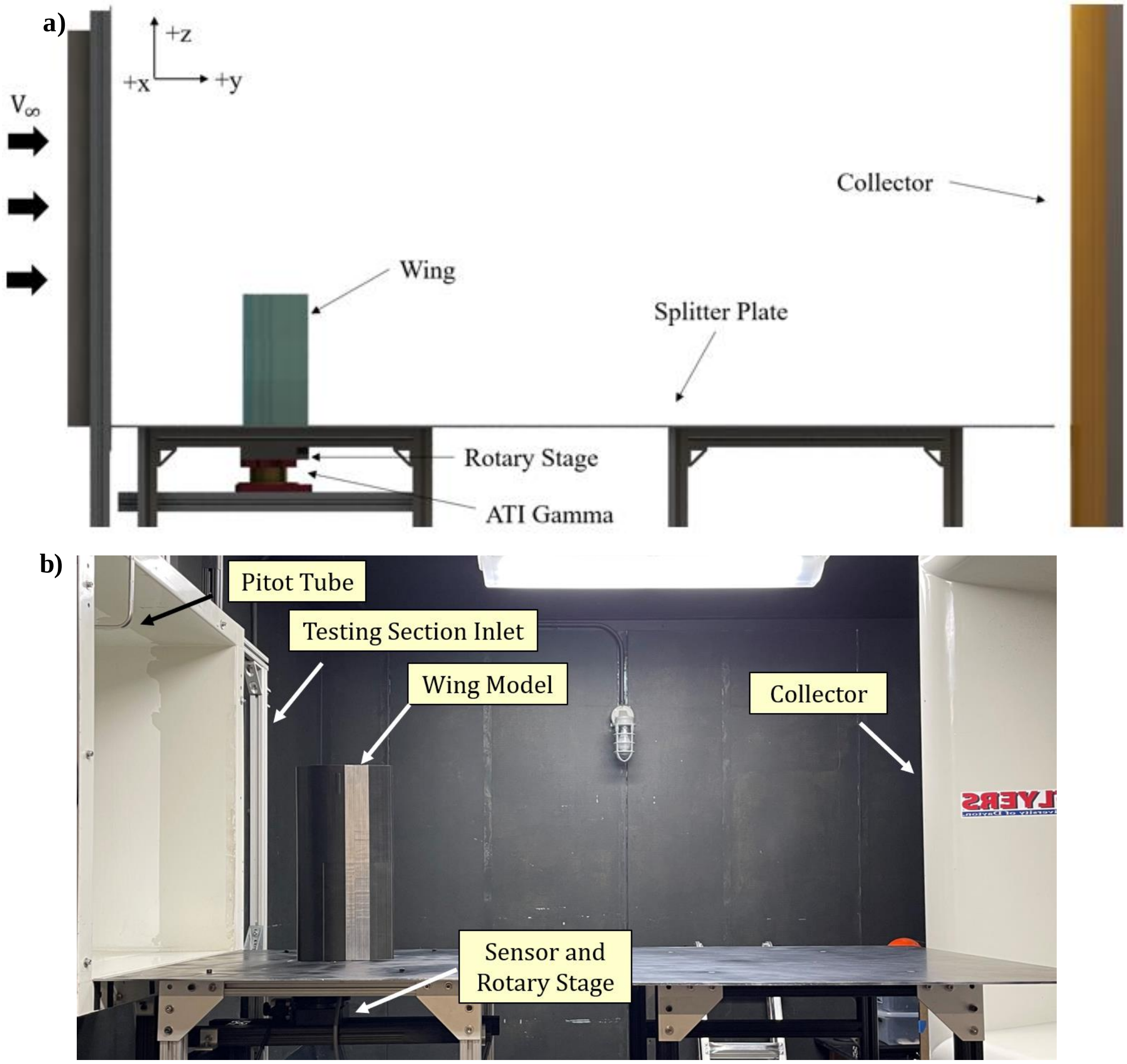

Figure 1. a) Rendered CAD Schematic and b) Picture of UD-LSWT Experimental Setup

Table 1. Test Matrix for Force Based Experiments

Wing Configuration	Reference Area(m^2)	Wing Reynolds Number	Wing angle of attack (degrees)	Freestream Velocity (m/s)	Effective Camber Deflection (degrees)
Trailing-Edge Device	0.0826	200,000 270,000	-15 to 15	15 20	0, 1.7, 3.0 4.6
Morphing Wing					0, 3.0, 5.1

To better compare the performance of the morphing wing, the performance of the Eppler 479 airfoil with a traditional trailing edge device with the same planform area was also tested. The hinge of the trailing edge device was located at 70% chord of the airfoil. The actual hinge as well as the gaps between the wing and the trailing edge device are not included in the design. Instead, different swappable trailing edge pieces were designed and was swapped for different flap deflection cases. Hence, the current study will focus on the aerodynamic performance comparison of the airfoil shape without the consideration of any additional interference. Four different wing trailing edge deflection angles: 0°, 5°, 10°, and 15° are tested in the current study, which has an equivalent δ_{eff} of 0°, 1.7°, 3.0°, and 4.6° respectively, which is shown in Figure 3.

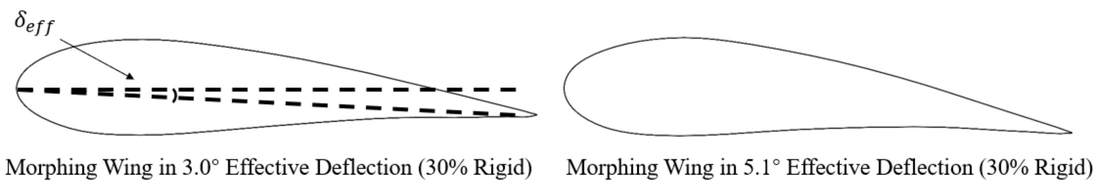

Figure 2. Schematic of the Different Morphing Airfoil Shapes with 30% Rigid Leading Edge

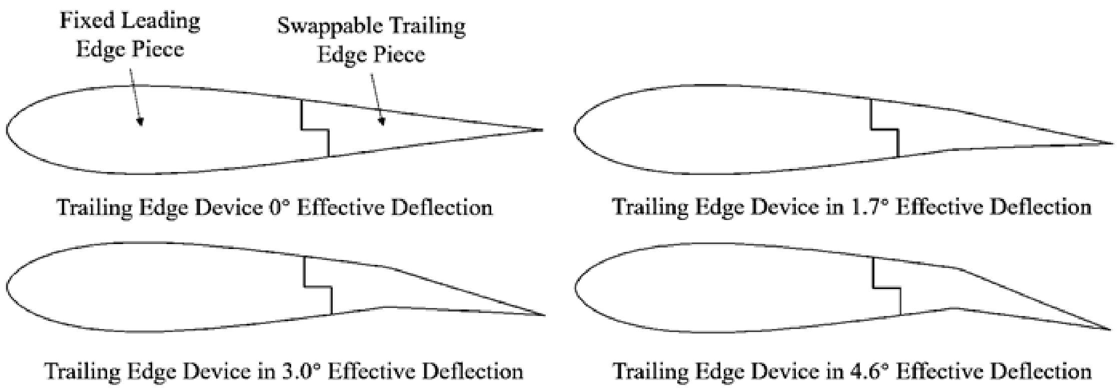

Figure 3. Schematic of the Different Traditional Trailing Edge Device Airfoil Shapes

B. Numerical Setup with FlightStream®

FlightStream® [14] was used to numerically investigate the same models tested in the wind tunnel to obtain the pressure distribution comparison between the traditional and the morphing wing. FlightStream® contains two main types of potential flow solvers: pressure-based solver and vorticity solver. The pressure-based solver is used in the present study, which utilizes the pressure fields on the model geometry to determine the aerodynamic loads. A combination of source and sink panels along with doublet panels on the trailing edge is used to provide a solution for pressure-based potential flow solvers.

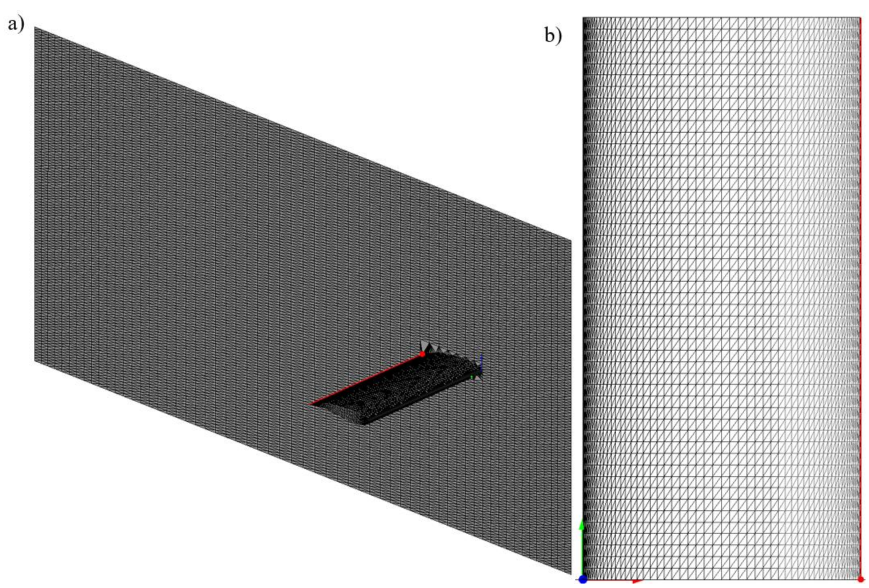

Figure 4: Surface Mesh in Flightstream for a) Isometric View of the Wing and Splitter Plate and b) Top View of the Wing

For the current study, the pressure-based solver is selected based on good agreement with the wind tunnel experimental result in our previous studies [17]. The boundary layer is assumed to be fully turbulent with the viscous coupling enabled.The surface mesh of the wing and the splitter plate in FlightStream® is shown in Figure 4. The wing platform has a structured mesh with 80 node points spanwise and 60 node points chordwise on both upper and lower surfaces. The chordwise mesh also has a dual side growth rate of 1.05 to better resolve the leading and trailing edge geometry. Moreover, the splitter plate of the same size as the one used wind tunnel experiment is included in the model to better represent the actual experimental conditions. The splitter plate has 60 node points from side to side and 80 node points, with a dual-side growth rate of 1.05 downstream.

III. Results

A. Experimental Results

The wind tunnel experimental results will be discussed in this section. The maximum variation between the different Reynolds number tested at 15 m/s and 20 m/s among all cases was less than 2% on lift coefficient, and 6% drag coefficient. As such, the experimental result from the 20m/s case (Re = 270,000) was chosen to be a good representative to show the overall aerodynamic performance of both wing configurations and discussed in detail in this section.

Experimental result on the lift coefficient (C_L) vs. angle of attack (α) is shown in Figure 5. The wing with trailing edge devices represents a traditional configuration and is shown in red markers. For the $\delta_{eff} = 0°$ case, which has the original Eppler 479 airfoil shape, C_L changes linearly with the increment in α within the α range tested as expected. Based on the McCormick formula [18] (Equation 1), the effective AR calculated based on the lift curve slope for the baseline wing is 2.90.

$$a = \frac{dC_L}{d\alpha} = a_0\left(\frac{AR}{AR + 2\left(\frac{AR+4}{AR+2}\right)}\right) \tag{1}$$

As δ_{eff} for the trailing edge devices increases, C_L is increased at all α tested. An average increment of 0.16 C_L is found when increasing the effective camber angle of 1.5°.

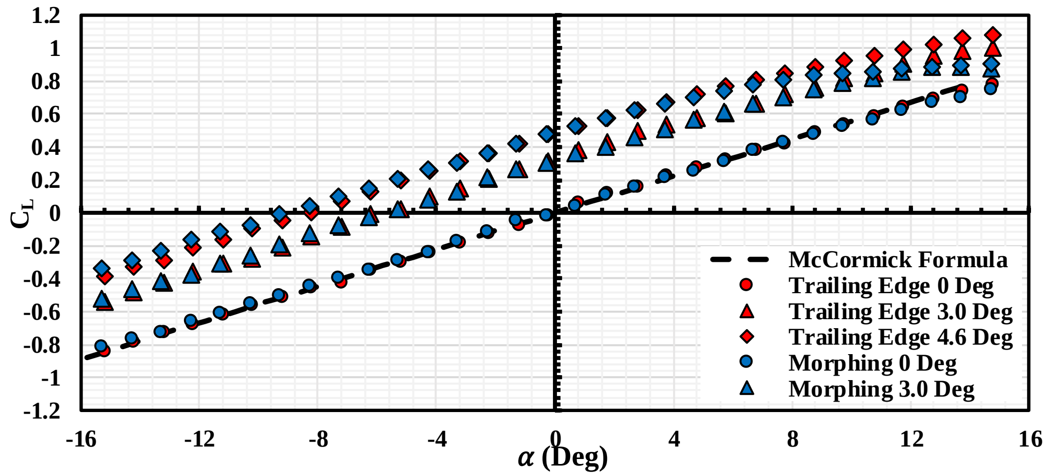

Figure 5. C_L vs. α for Trailing Edge Device Wing and Morphing Wing for all Deflection Angles Tested

The experimental result for the morphing wing is marked in blue. For the $\delta_{eff} = 0°$ morphing wing case, the experimental results overlap on the $\delta_{eff} = 0°$ case for the trailing edge device wing. When the morphing wing actuates, the C_L for $\delta_{eff} = 3.0°$ overlaps on the results for the 3.0° trailing edge devices case, indicating a similar lift generation capability when having a similar effective camber angle. Further increase in the morphing angle results in a higher increment in C_L, where the results for the $\delta_{eff} = 5.1°$ morphing wing case overlaps on the $\delta_{eff} = 4.6°$ trailing edge devices case at small α. However, at high α, the morphing wing beings to perform worse and stall at α~12°. This is likely due to flow separation occurring at higher morphing angles. Further investigation of this phenomenon will be discussed with the Flightstream simulation results in the next section.

Figure 6 shows the drag coefficient (C_D) vs. α for all cases in Figure 5. As the morphing wing is manufactured with a flexible material, the surface finishing is different from the traditional trailing edge wing model. As the airfoil shape and the C_L performance for both $\delta_{eff} = 0°$ cases are identical, a skin friction drag correction is applied to the morphing wing cases intending to minimize the effect of different skin friction. To do this, the difference in the C_{D_0} between the morphing wing and the traditional wing at 0° deflection is subtracted from all morphing wing cases at all α tested. The resulting C_D value is shown in Figure 6.

Similar to the C_L results, C_D performance for both $\delta_{eff} = 0°$ cases overlap each other. When increasing δ_{eff}, C_{D_0} is shifted to a lower α due to the change in $\alpha_{c_L=0}$ for all cases, as expected. At higher δ_{eff}, the morphing wing produces a lower drag at $-10 < \alpha < 10$ when compared to its traditional counterpart, while higher C_D is observed at high α due to stall. Additionally, the difference in C_D between the trailing edge device wing and the morphing wing becomes more pronounced at the highest δ_{eff} cases, indicating better performance for the morphing wing at higher lift conditions.

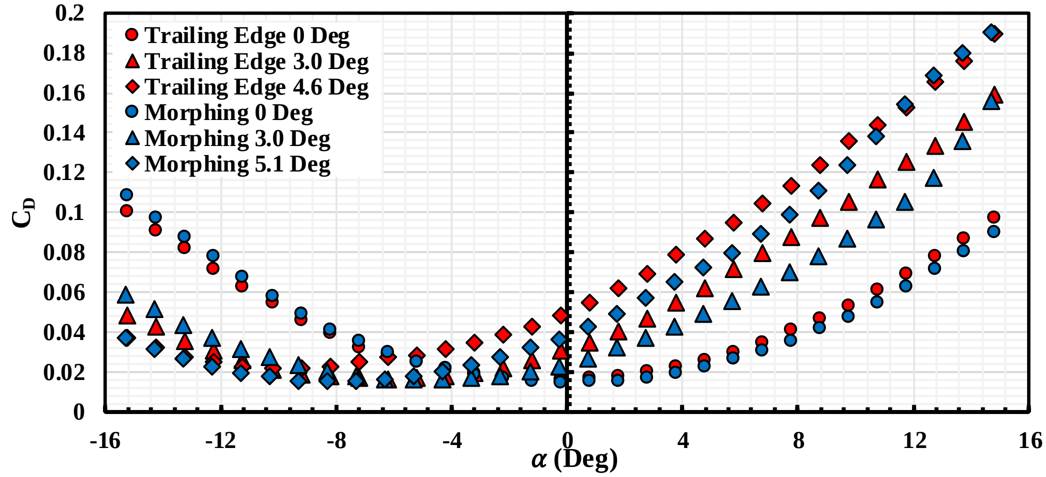

Figure 6. C_D vs. α for Trailing Edge Device Wing and Morphing Wing for all Deflection Angles Tested

The drag polar for both wing cases is shown in Figure 7. At $C_L = 0$, the morphing wing has a relative constant C_D and C_{D_0} at each δ_{eff}, while the trailing edge device wing has an increment in C_D for a higher δ_{eff}. This indicates a better overall performance for the morphing wing when compared to its traditional counterpart. Moreover, at $0 < C_L < 0.8$, the morphing wing shows a lower C_D when generating the same C_L while a higher C_D is observed for the morphing wing at negative C_L range. It is worth noting that since the morphing wing mechanism serves as a trailing edge device, it would be less likely to operate in a condition where $C_L < 0$.

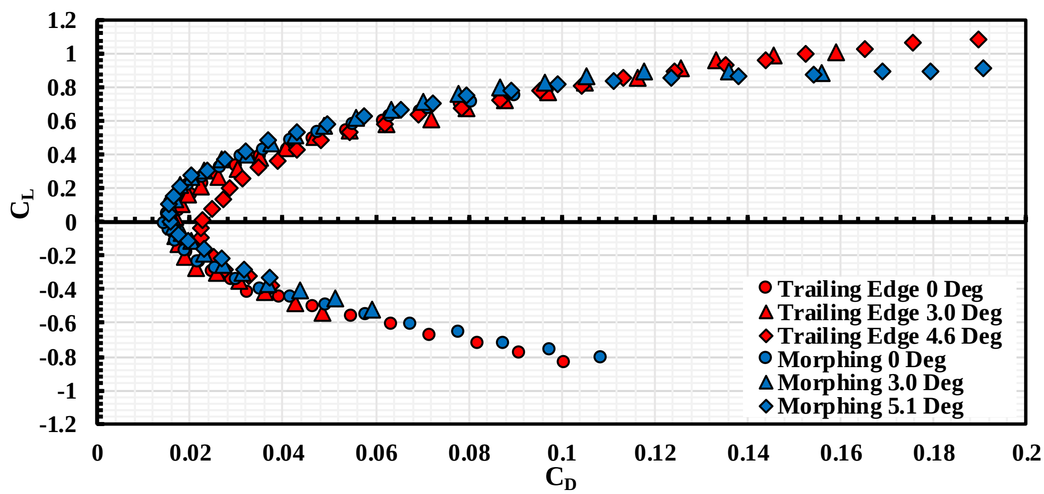

Figure 7. Drag Polar for Trailing Edge Device Wing and Morphing Wing for all Deflection Angles Tested

IV. Numerical Analysis Results

A. Comparison of Experimental and Numerical Simulation Results

The experimental and numerical comparison of C_L is shown in Figure 8.a for the traditional wing and 8b for the morphing wing. The experimental results are shown as discrete data points while the Flightstream results are shown as lines. For all trailing edge cases, results from Flightstream matches the experimental results well, except for higher trailing edge deflection cases at negative α where Flightstream predicts a smaller negative stall angle when compared to the experimental results. For the morphing wing cases, the results from Flightstream also matched the experimental results, except for the 3° morphing wing case where Flightstream overpredicts C_L at small α ranges.

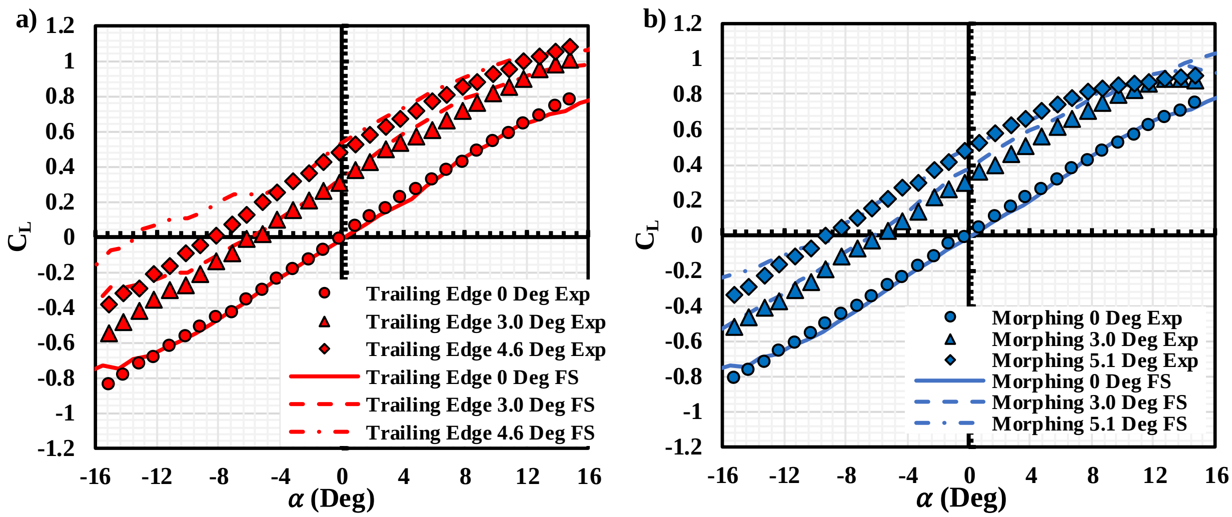

Figure 8. C_L Comparison Between the Experimental and Flightstream Results for a) Trailing Edge Devices and b) Morphing Wing

The comparison of C_D is shown in Figure 9. For both cases, Flightstream over predicts C_D, especially at negative α range. At positive α, Flightstream overpredicts the drag for the trailing edge cases by ~5% while overpredicts the morphing wing cases by ~10% between $-10° < \alpha < 15°$. Overall, Flightstream was able to estimate the aerodynamic performance for both cases, which provides confidence for further numerical analysis of other morphing wing designs.

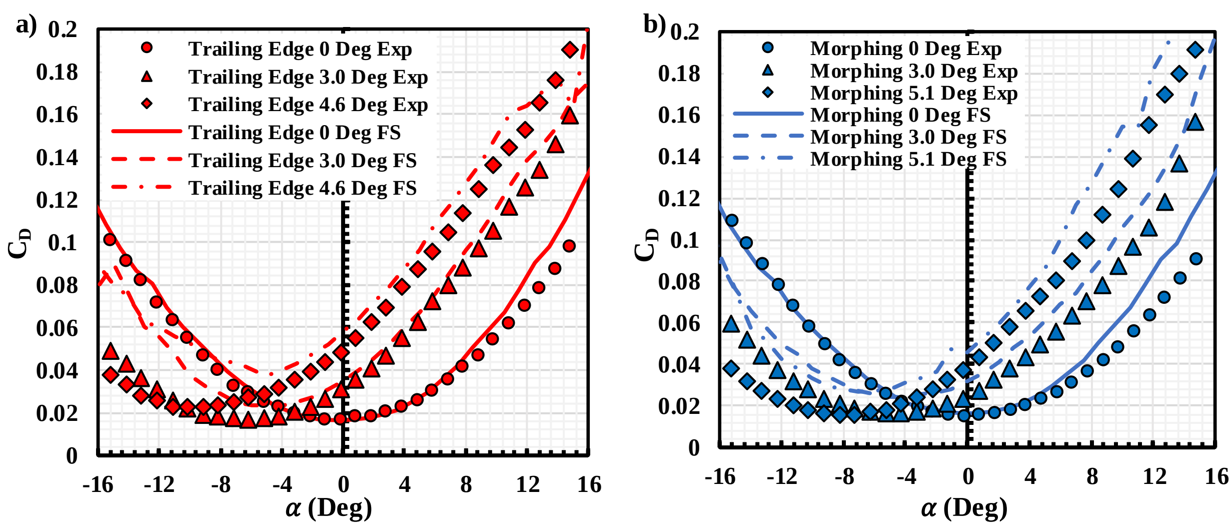

Figure 9. C_D Comparison Between the Experimental and Flightstream Results for a) Trailing Edge Devices and b) Morphing Wing

C. Pressure Coefficient Contour

The surface pressure coefficient (C_P) contour for the trailing edge device case at 4.6° deflection, and the morphing wing (30% rigid leading edge) at 5.1° deflection at $\alpha = -5°$, 0° and 5° are shown in Figure 10. A clear difference in the pressure distribution can be observed between both cases. For the trailing edge device, a sudden negative pressure coefficient is observed at the hinge location in the upper surface due to the abrupt change in the curvature of the wing. However, no sudden change of pressure is observed in the morphing wing case at all angles of attack due to continuous morphing of the trailing edge. Also, due to the continuous morphing, an increment in positive pressure coefficient occurs closer to the mid-chord region , indicating an earlier flow separation when compared to the traditional wing. An increased negative C_P is also observed at the leading edge of the morphing wing when compared to the trailing edge device indicating an increase in suction. This is likely due to the higher curvature of the airfoil caused by the morphing.

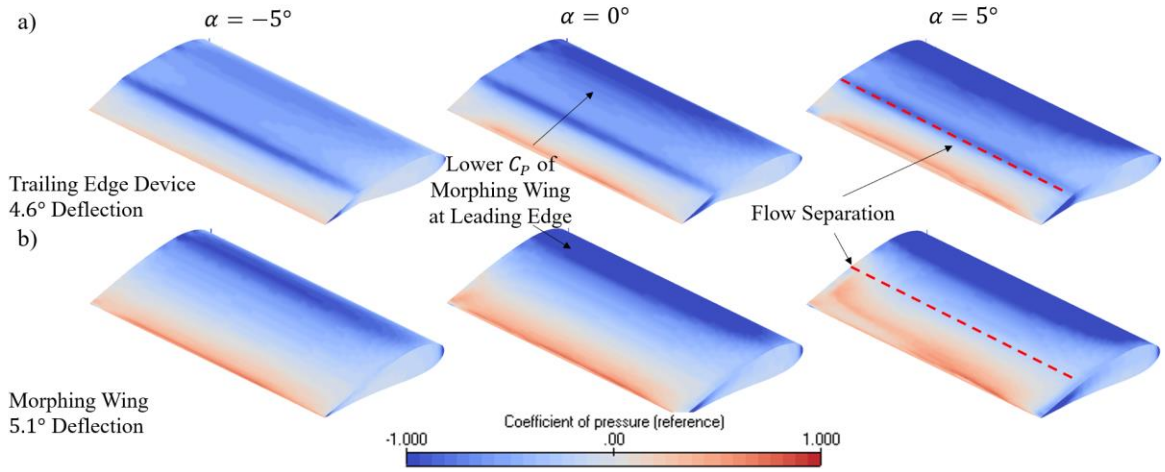

Figure 10. Surface Contour of Pressure Coefficient (C_P) for a) Trailing Edge Device at 4.6° Deflection and b) Morphing Wing (30% Rigid) at 5.1° Deflection at $\alpha = -5°, 0°$ and 5°

The sectional C_P contour for both wing configurations in Figure 10 at $\alpha = 0°$ and $\alpha = 5°$ is shown in Figure 11. It is clear that a sudden change in C_P is observed for the trailing device wing at 75% chord location at both the upper and lower surface of the airfoil, where the hinge point of the trailing edge device is located. A lower C_P is observed at upper surface and higher C_P at lower surface of the airfoil at this location, resulting in a higher lift generation. This is also observed in the C_P contour in Figure 10. While for the morphing wing, a lower peak C_P is observed closer to the leading edge, resulting a higher lift generation. However, a drop in C_P is also observed after 35% chord indicating a possible flow separation occur after this location, especially for $\alpha = 5°$ case.

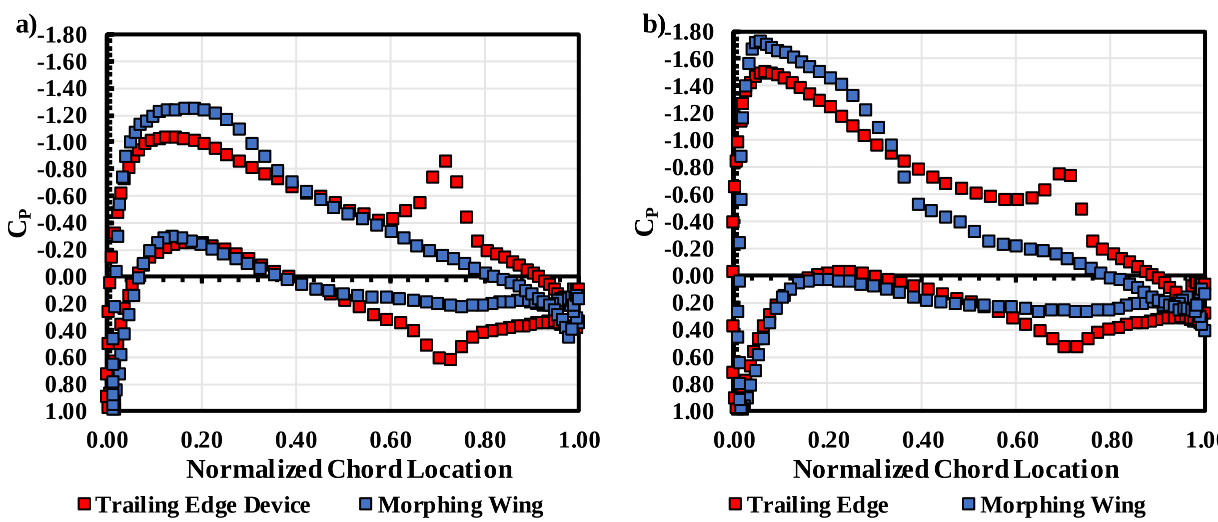

Figure 11. Sectional C_P Contour of Both Wing Configurations in Figure 10 at Mid-span for a) $\alpha = 0°$ and b) $\alpha = 5°$

V. Conclusion

The novel morphing wing and its traditional counterpart using a trailing edge device were investigated experimentally and numerically. With a simple rigid-flexible mechanism, the morphing wing was able to achieve a similar lift performance as the traditional trailing edge device wing at the same effective camber deflection angle. At an angle of attack between -10° and 10°, the morphing wing also generates less drag when achieving the same lift performance, leading to higher aerodynamic efficiency for the morphing wing. However, the morphing wing experiences an earlier stall when compared to its traditional counterpart at higher effective deflection angles.

Flightstream numerical analysis agreed well with the wind tunnel experimental result on lift generation while overpredicting the drag forces. The surface pressure contour revealed a sudden change of the pressure for the morphing wing when deflected at higher effective camber angles, indicating a possible flow separation leading to the early stall phenomena observed in the wind tunnel experiment. A possible solution for the second generation of morphing wing design intending to prevent an early flow separation through changes in the morphing mechanism.

VI. Future Work

Based on the wind tunnel experiment and the Flightstream analysis on the current morphing wing design, two additional morphing wings with the same morphing structure are designed. The second-generation morphing wing has a larger leading edge rigid section intending to delay the chord location where the airfoil shape morphs and delays the flow separation at a low α range. The scaled prototype with a span of 0.076m and a chord of 0.203m was built, and the morphing airfoil shape is acquired with the prototype and is shown below in Figure 12. Further Flightstream analysis will be conducted and compared the performance with the current design. With promising analysis results, the wind tunnel experiment will be conducted to future investigate the aerodynamic performance of this novel morphing wing concept.

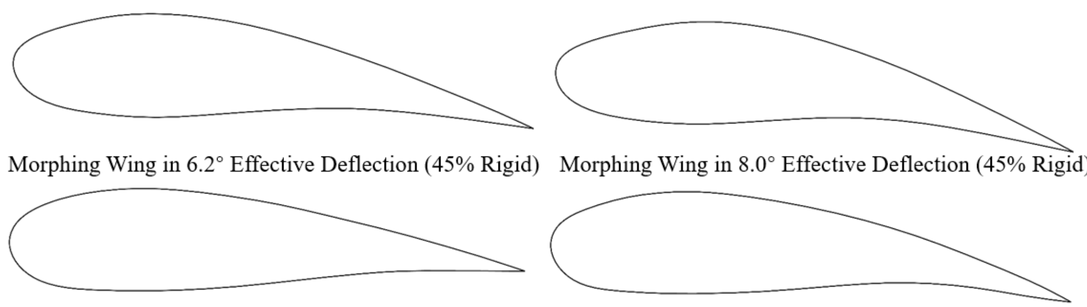

Figure 12. Schematic of the Different Morphing Airfoil Shapes with 45% and 55% Rigid Leading Edge

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